Deep Learning-Based GNSS Clock Bias Prediction

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***Abstract*—Clock Bias is one of the most fundamental problems that causes errors in Global Navigation Satellite System (GNSS) positioning. Predicting these errors is crucial for improving the GNSS navigation precision. We used the actual GNSS satellite clock data by Crustal Dynamics Data Information System (CDDIS) to assess how the different Recurrent Neural Network (RNN) architectures varies while predicting the satellite clock-bias.We trained Gated Recurrent Unit Networks (GRU) and Long Short-Term Memory (LSTM) models and found that GRUs handled the raw drifting bias more accurately.GRUs obtained a R^2 score of 0.98 which was much better than that of LSTMs(0.05-0.49).Subsequently, we used a basic trends-residual transformation for the satellite bias signal, which helped LSTMs to achieve R^2 of 0.99.We also observed that the interpolation of the clock drift affected the forecasting performance notably.Finally, we combined all these protocols and presented a clean and replicable pipeline for the GNSS clock bias prediction.**

***Index Terms*—Deep Learning, Clock Bias Prediction, LSTM, GRU.**

I. Introduction

[1] Global Navigation Satellite Systems(GNSS) transformed the navigation and timing services across various fields such as telecommunication, more precised maps, etc. The foundational accuracy of the GNSS depends upon the 2 crucial factors: the clock bias and the ephemeris data.Satellite clock bias serve as divergence of the atomic clocks from system provided by system whereas broadcast ephemeris gives the orbital parameters which are important for computing the satellite positioning.Even the trivial errors in these can substantially decrease the accuracy of the positioning of the satellite, specially in predictions involving the centimeter-level precision being [2] Precise Point Positioning(PPP) and the real-time kinematic(RTK) positioning.

Conventional procedures for predicting the satellite clock bias ephemeris errors have been mainly relied on the polynomial models and the physics-based techniques that are effective to a certain point but later fails to catch the complex and nonlinear flux that is influenced by the various factors such as temperature, solar radiations and relativistic dynamics [3]. GNSS constellations keeps on increasing so the user demands for for higher accuracy, all of these traditional methods are appeared to have shown limitations and restrictions in handling the satellite data.The occurrence of the Machine Learning and the Deep Learning technologies have proposed brand- new approached for handling these challenges, offering a methodology that works with [4], [5] linear and non linear data and can learn various patters that are rather very difficult for traditional methods to match. Recently,there is a upsurge in applying various Deep Learning and Artificial Intelligence techniques for predicting the GNSS error clock bias and ephemeris predictions. Most of these models are Recurrent Neural Networks(RNNs) such as LSTMs [6] and GRUs [7] which works exceptionally good on long-term depending sequencing data. These approaches takes huge amounts of data to learn the clock bias patterns and the ephemeris error dynamics, making the predictions more accurate readjust for the particular satellite attribute accordingly including constellations like Galileo [4], [8], BeiDou [9], GPS. However, use of these clock bias errors has to be carefully modeled as the raw satellite clock signals contains strong drift characteristics that shows the challenges that different RNN architectures handle at varying degrees of accomplishment. This study introduces two different modeling strategies for the high- precision satellite clock bias prediction using the actual GNSS data from the Crustal Dynamics Information Systems(CDDIS). [10]The coefficient of determination (R^2 metric was used as the evaluation metric. We found that GRUs performed much better initially and handled raw drifting bias signal efficiently, achieving a R^2 of 0.98, whereas LSTMs performed poorly compared to GRUs on untransformed data (R^2 ranging from 0.05 to 0.49). To resolve this issue, we proposed a approach of trend-residual decomposition that transformed the bias signal to it's deterministic trend and strong residual components enabling LSTM model to achieve a R^2 of approximately 0.99. It was revealed that interpolation selections for the clock drift significantly affect the prediction accuracy.

II. Literature Review

GNSS have transformed into a prime preference for the precision positioning and the timing of the applications globally,with their accuracy mainly depending upon the precision of the satellite clock bias prediction and the broadcasting of ephemeris data. Conventional practices for managing of these parameters have gradually been substituted with the Machine Learning approaches, specifically Deep Learning, which illustrates superior performance in complex temporal dependencies. [3]Long Short-term Memory(LSTM) architectures have established themselves as a strong approach for the satellite clock bias prediction because of their powerful ability to seize the long-term time-based dependencies in the time-series data [5]. Huang et al. introduced that a supervised LSTM that was specifically designed for the satellite clock bias prediction, explaining the importance of this architecture in the handling of the sequenced patterns that are present in the satellite clock. Expanding on the foundational work done on Gated Neural Networks (GRUs) by Chung et al. [7], the LSTM architectures states the vanishing gradient issue in the traditional recurrent neural networks using specialized gating procedure the regulated the flow of information, enabling the neural network to selectively preserve or forget the information over elongated periods of times. Bhatt et al. [6] then explored the applications of the LSTM for prediction of satellite clock bias, reinforcing the ability of the architecture to learn complex time-based patterns from the previously recorded clock data and efficiently modeled both short-term and the long-term variations in drift patterns in satellite clocks.
[8]The increasingly usage toward hybrid models is a approach that is discussed by Cai et el., who proposed a IF-TEA- LSTM model for deviation detection in BeiDou Navigation System(BDS),achieving a 68.37% enhancement in the clock bias deviation accuracy for the satellites MEO for the short-term prediction and minimizing the root mean square error (RMSE) upto 20.80%. This hybrid approach stated the unique characteristics of the different satellite types (GEO,IGSO and MEO) and effectively improved detection strength using data integrity evaluations and the stationarity assessments.

The comprehensive study by Lv et al. [3] compiles a research on the satellite clock bias prediction models in various GNSS constellations,contributing a important and unique perspective on the evolution of conventional parametric ap- proaches to new machine learning methods and stating the importance of this new shift from only physics-based models to data-driven approaches that can acquire complex, temporal, nonlinear relationships in the satellite clock patterns. The amount and precision of broadcast ephemeris data directly influences the positional accuracy addressing necessisity of more robust quality checking methods. Alonso et al. [11] deeply analyzed Galileo F/NAV ephemeris errors and the broadcast clock data utilizing 43 months of data(January 2017 to July 2020), implementing a automated monitoring the navigation data for detection and verification potential errors the satellite broadcast navigation data. Lapin et al. [12] analysis revealed that the RINEX files do not give enough information to access CCE parameters sent at every time instance intervals, addressing the importance for the use of binary navigation data from global monitoring networks, and found out that maximum TTCS is affected by use of data (IOD) updation periods and structures with unmatched IOD parameters,potentially introducing extended delays to time to first fix(TTFF). These finding have a necessary implications for real-time positioning application and the satellite-based augmentation systems(SBAS) message timing requirements.

The transformer architecture introduced by Vaswani et al. [13] (2023) in ”Attention is All You Need” is a historic benchmark in Deep Learning and has revolutionaized sequence modeling across various domains, while it is still not applied extensively for satellite clock bias prediction, the study promises a great potential for capturing the temporal relationships in the GNSS data through the self-attention mechanism that can evaluate the importance of different time stamps.With the integration of the advanced neural network architectures and comprehensive feature engineering representing the most promising path forward for next-generation satellite navigation systems.

TABLE I
Variable Description in the Processed Dataset

Variable	Description
timestamp	UTC time extracted from the `.clk` records.
bias_float64	Satellite clock bias converted to 64-bit floating point.
drift_float64	Satellite clock drift (rate of change of bias).
t_seconds	Time converted to seconds since start of day.
drift_final	Interpolated continuous drift series (5-minute gaps removed).

III. Methodology

A. Dataset Description

The dataset we used in this study is from the **NASA Crustal Dynamics Data Information System (CDDIS)**, which provides precise International GNSS Service (IGS) clock solution files in **.clk** format. These files contain high-precision satellite clock bias generated at 30-second intervals using global GNSS tracking networks.

For this study, we focus on the satellite labeled *C12* under the *AS* category. This satellite was chosen because its clock bias series was complete for the days (2021, January 14–20), contained no structural discontinuities.

The raw `.clk` files contain no explicit time-in-seconds column, so it was computed manually based on record order. The data fields used in this work are summarized in Table I.

B. Overall Workflow Overview

To give a clear picture of our entire pipeline, Fig. 1 summarises the complete flow used in this study. The workflow highlights both approaches explored in this work: the direct modelling of raw clock bias using GRUs, and the trend–residual based method where an LSTM is applied after separating long-term trends from the bias signal.

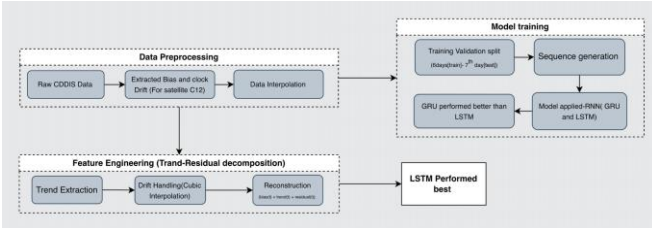

Fig. 1. Overall workflow of the proposed satellite clock bias forecasting pipeline.

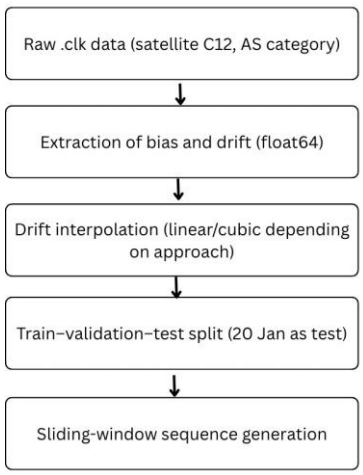

Fig. 2. Overview of the preprocessing pipeline applied before model-specific steps.

The diagram outlines how the raw CDDIS clock data were processed, how sequences were generated for training, and highlights our research approach in a basic view.

C. Preprocessing

The preprocessing stage was kept deliberately identical for both approaches to ensure that performance comparisons remained fair and consistent. Although the two modeling strategies diverge later in how they treat the bias and drift components, the foundational data preparation steps were shared across all experiments. These steps include cleaning, feature construction, normalization, train–test splitting, and sliding-window sequence formation.

1) Cleaning and Alignment: The International GNSS Service (IGS) `.clk` files provide satellite clock corrections at a 30-second sampling interval. Our dataset contained multiple satellites across the AR and AS categories; this work focuses on the C12 satellite under the AS category. Only two variables were retained: *clock bias* (seconds) and *clock drift* (seconds/second), both stored in `float64` precision. The timestamp column was standardized to UTC and sorted chronologically

Since the original `.clk` files do not provide an elapsed-time field, we computed a time-in-seconds variable.

2) Drift Continuity via Interpolation: The drift values in the raw dataset are sampled every 5 minutes, creating uniform gaps. In order to obtain drift values consistent and uniform, interpolation was performed. Although each modeling approach later uses a different interpolation mode (linear for the GRU model and cubic for the LSTM-residual model):

$$\tilde{d}(t) = \mathrm{interp}(t; t_i, d_i),$$

where (t_i, d_i) are the observed drift samples.

This resulted in uniformly sampled drift series, referred to as `drift_final`.

3) Feature Construction: After alignment, three fundamental variables were available for both modeling pipelines:

- **bias_float64**: the raw satellite clock bias.
- **drift_final**: interpolated continuous drift.
- **t_seconds**: elapsed time since start of Jan 14.

These serve as the base feature set, after which each approach applies its own additional transformations (e.g., trend–residual decomposition) described later in their respective subsections.

4) Normalization: To ensure numerical stability and faster convergence during training, all features were scaled using a MinMaxScaler fitted only on the training partition for the LSTM model. Whereas for GRU, StandardScaler was used. Scaling was performed independently for each feature to prevent unintended correlations. All scalers were saved for use later during prediction and in the deployment.

5) Train–Test Split: The training window consisted of data from *January 14 to January 19*. The target prediction day, January 20, was held out entirely for evaluation. This ensured that all models were tested on an unseen sequence. The split respected chronological order:

$$D_{\text{train}} = \{t \in [14\text{Jan}, 19\text{Jan}]\}, \quad D_{\text{test}} = \{t = 20\text{Jan}\}.$$

6) Sliding-Window Sequence Generation: The models were trained using a one-step-ahead prediction strategy. Given a historical window of length L, the input sequence is

$$X = \{x_{t-L+1}, \ldots, x_t\},$$

and the target is the next-step value x_{t+1}. We used $L = 60$ for the GRU model (Approach 1) and $L = 10$ for the trend–residual LSTM (Approach 2), following the experimental design described in each section. This windowing method enables the model to learn short-term dependencies, and aligns with the design philosophy of both GRU and LSTM architectures used in the later sections.

D. Approach 1: GRU Model on Raw Bias

1) Motivation: GRU or Gated Recurrent Unit are a type of recurrent neural network and they are well suited for time series data with smooth monotonic trends that is a consistent increase of decrease in variable over time and as we saw earlier that satellite clock bias drifts slowly and has long term temporal dependencies. GRU are really computationally efficient for long sequences and they handle vanishing gradients

better on monotonic signals/trends. clock bias data is really precise and sensitive and GRU gates help preserve long range dependencies without losing subtle variations in as low as picosecond level bias which was important for this research.

2) Model Inputs: We took two features into consideration, clock bias which was converted to `float64` as the primary input and `drift_float64` as the secondary feature, GNSS satellite bias values are extremely small typically in the range of 10^{-4} to 10^{-8} seconds and single precision `float32` is not enough and would cause quantization errors, harming both training and evaluation. Clock drift was linearly interpolated to repair missing values and this is not a issue as drift is a derived, smoothly varying quantity. Drift was included because it physically influences bias evolution and a sampling rate of 30 sec reindex ensures strict temporal regularity and standardscaler was applied independently to bias and drift. The window size was 60 samples (30 minutes of past data used to predict next value).

3) Architecture: The proposed forecasting model consists of a stacked Gated Recurrent Unit (GRU) neural network structure, which is intended to reflect the signal bias of the GNSS system over either a short-term or long-term scale. The proposed model architecture initiates with a 128-unit GRU layer, which is designed to retain the whole sequence output. This allows the proposed model to infer temporal characteristics over the whole 60-step input sequence. This is followed by another GRU layer, with 64 units, which translates the whole temporally extracted feature into a compact representation. In order to overcome overfitting in the high-precision bias signal, which has less variance, a dropout layer with a dropout rate of 0.2 is introduced. The extracted features are then transferred to another 32-unit fully connected layer with ReLU activation. The final output is achieved using an additional Dense(1) layer to predict the next step of the clock bias signal.

The model has been trained using the ADAM optimizer with the mean squared error loss function. In order to prevent overfitting, the last 10% of the training set has been set aside as the validation data, using EarlyStopping. During training, the batch size has been set to 64, using ModelCheckpoint in order to keep track of the best model weights.

In ”Training Deep Neural Networks,” it’s noted that ”in practice, the number of training iterations until convergence takes anywhere from 50-100 epochs, depending on the values randomly assigned to the model’s weights.”

Our final architecture consists of two stacked GRU layers (128 and 64 units), followed by ReLU-activated dense layers, enabling the network to extract both long-term and short-term temporal dependencies.

TABLE II
GRU MODEL ARCHITECTURE

Layer	Configuration
GRU Layer 1	128 units, return sequences=True
GRU Layer 2	64 units
Dropout	0.2
Dense Layer	32 units, ReLU
Output Layer	Dense(1)
Loss	MSE
Optimizer	Adam
Batch Size	64

4) Output: The most important thing we focused was getting picosecond level prediction accuracy as even extremely small errors in satellite clock bias propagate directly into user range measurements. A timing error of just 1 nanosecond translates to approximately 30 centimetres of range error. Forecasting the satellite clock bias with picosecond resolution is not merely desirable, but necessary to preserve the accuracy and reliability of downstream GNSS applications. Therefore, maintaining and forecasting the clock bias at the picosecond scale is necessary to support high-precision GNSS positioning and timing services.

The GRU model produces a one-step-ahead prediction of the satellite clock bias for the next 30-second epoch. After generation, the predicted values are inverse-transformed using the bias scaler to keep the data in original float64 precision format. Model performance is evaluated exclusively on the test seventh day of data to ensure that the prediction accuracy reflects genuine generalization to unseen temporal segments.

Standard regression metrics include the absolute error, root-mean-square error, coefficient of determination, and Pearson correlation. The model achieves an MAE of roughly 2.9×10^{-8} seconds and an RMSE of about 3.5×10^{-8} seconds, corresponding to picosecond-level predictive accuracy. A high score in R^2 value close to 0.98 and a Pearson correlation coefficient greater than 0.999 show that the predicted bias values are very well aligned in time with the true values. This means that the GRU really succeeded in catching both the long-term drift and the fine-scale variations within the satellite clock signal.

E. Approach 2: LSTM with Trend–Residual Decomposition

This approach represents our best-performing model. The core idea is to simplify the learning problem by separating the long-term linear behaviour (trend) from the short-term stochastic variations (residual). By training the LSTM on the residual component instead of the raw bias directly, the sequence becomes significantly easier as well as much more relevant for the model to learn.

1) Motivation: LSTMs are powerful sequence models, but they often struggle with long, slowly varying linear trends. When such trends dominate the signal, the network tends to allocate its capacity to fit the trend instead of learning the short-term temporal structure. To address this, the bias time series was decomposed into:

- a linear trend component, and

- a residual component that captures the rapid short-term variations.

2) Trend Extraction: The trend was extracted using a simple least-squares linear fit over the full bias sequence. For each time index t, the trend is computed as:

$$\text{trend}(t) = at + b,$$

where a and b are obtained from a first-order polynomial fit:

$$[a,\ b] = \text{polyfit}(t,\ \text{bias},\ 1).$$

The residual series is then defined as:

$$\text{residual}(t) = \text{bias}(t) - \text{trend}(t).$$

This ensures that the LSTM focuses exclusively on learning the short-term fluctuations.

3) Drift Handling: The data for drift variable was only available at 5 minute intervals compared to that for clock bias which was available for every 30 seconds. Hence, the missing drift values were interpolated using *cubic interpolation*, which produced a smoother sequence than linear interpolation. Since the LSTM is trained on the residual (a smooth, stationary signal), using a smoother drift input would provide more stable learning compared to the linear interpolation used in the GRU approach.

4) Model Inputs: The LSTM receives a three-element representation at each time step:

- the residual component of the clock bias,
- the cubic-interpolated drift value,
- the time index t_{seconds}.

The residual and drift values are scaled with Min–Max normalization before being passed to the network.

5) Output: The model is trained to predict:

- the next-step drift value, and
- the next-step residual value.

6) Reconstruction: After predicting the next residual, the clock bias prediction is reconstructed by re-adding the linear trend:

$$\widehat{\text{bias}}(t) = \text{trend}(t) + \widehat{\text{residual}}(t)$$

Although simple, this decomposition makes the prediction problem significantly easier and leads to highly accurate bias estimates. This approach achieves an R^2 score of approximately 0.99 for the test set and models the raw clock bias which was given in the dataset.

IV. Results

This section presents the forecasting performance of both modelling strategies: (1) the GRU trained directly on the raw clock bias, and (2) the LSTM trained on residual and drift components after trend removal. All evaluations are carried out on the held–out day, 20 January 2024. The aim is to understand how well each model captures both the long–term evolution and the fine-scale variations in the satellite clock signal.

A. Approach 1: GRU on Raw Clock Bias

Before evaluating the model, the behaviour of the signal over the full week was inspected visually. Fig. 3 shows the clock bias from 14–20 January, which displays a clear linear- like drift with subtle, high-precision fluctuations superimposed. The corresponding drift values are shown in Fig. 4.

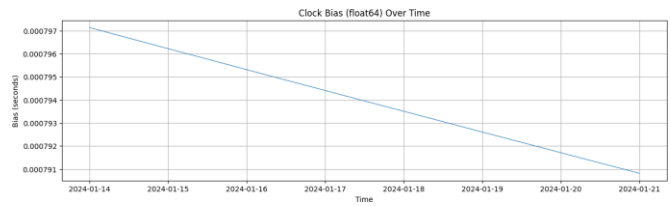

Fig. 3. Clock bias over time for 14–20 January (raw dataset).

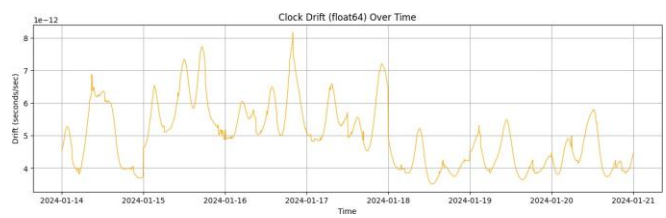

Fig. 4. Clock drift over time for 14–20 January (raw dataset).

These graphs show the smooth, monotonic nature of raw clock bias, which initially suggested the GRU could perform well. The GRU was trained on a 60-sample window (30 minutes) and evaluated on the full test day. Fig. 5 shows the predicted and true bias values for 20 January. The model follows the overall bias evolution closely, with minor deviations at some points of rapid change.

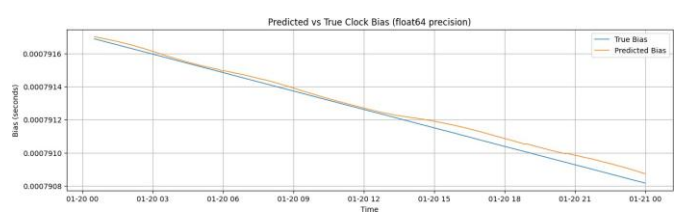

Fig. 5. Predicted vs. true clock bias on 20 January (GRU).

Short-term error behaviour is shown in Fig. 6 (prediction error over time) and Fig. 7, where a rolling MAE (window=200 samples) is plotted. The error does not remain constant throughout the period; instead, it follows a clear time–dependent pattern.

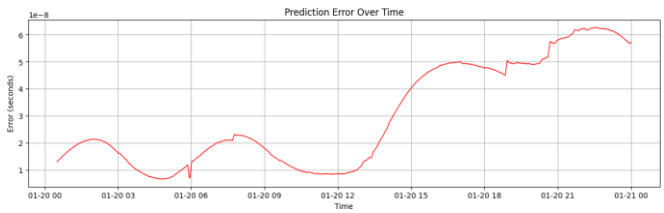

Fig. 6. Prediction error over time on 20 January (GRU).

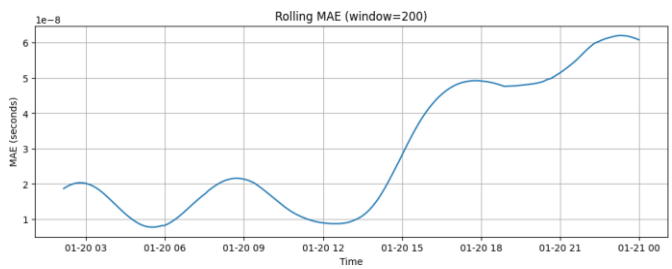

Fig. 7. Rolling MAE (window=200) on 20 January (GRU).

TABLE III
GRU PERFORMANCE ON 20 JANUARY

Metric	Value
MAE	2.99×10^{-8} s
MSE	1.26×10^{-15}
RMSE	3.55×10^{-8} s
R^2	0.9801
r	0.9991

The numerical results are summarised in Table III. The GRU achieves picosecond-level accuracy, and an R^2 of approximately 0.98.

B. Approach 2: LSTM with Trend–Residual Decomposition

The second approach separates the bias into trend and residual components. The LSTM is trained to predict the residual and the drift, while the bias is reconstructed using

$$\widehat{b(t)} = \widehat{trend} + \widehat{residual}(t)$$

This approach significantly reduces the burden on the model, since the residual sequence is much more stationary compared to the raw bias.

Fig. 8 shows the one-step drift prediction for the test day. The model closely matches the true drift curve, indicating good short-term learning.

Fig. 9 shows the reconstructed bias (trend + predicted residual). The alignment between the curves is very close, with almost no visually distinguishable error.

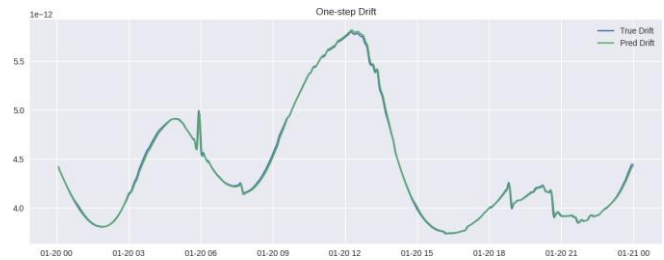

Fig. 8. True vs. predicted drift (one-step) for 20 January (LSTM).

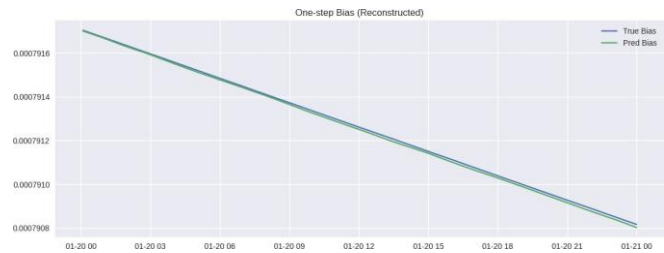

Fig. 9. True vs. reconstructed predicted bias (one-step) for 20 January (LSTM).

TABLE IV
LSTM RESIDUAL MODEL PERFORMANCE ON 20 JANUARY

Metric	Value
MAE	8.85×10^{-9} s
MSE	8.63×10^{-17}
RMSE	9.29×10^{-9} s
MAPE	0.0011%
R^2	0.9987

Table IV summarises the evaluation metrics. The LSTM achieves an R^2 of 0.9987, which is more than what the GRU model trained directly on the raw bias achieved.

C. Comparison of the Two Approaches

Both models perform well, achieving picosecond-level accuracy. However, the LSTM with trend–residual decomposition consistently outperforms the GRU in every metric. Note: We also tried LSTM with raw-clock bias as input and under identical conditions/hyper-parameters as the GRU model, however it could only achieve a maximum R^2 score of 0.49. Hence, Trend residual decomposition was used to increase its R^2 score to 0.99.

Removing the trend makes the residual sequence easier to model, and this translates into:

- lower MAE and RMSE,
- smoother one-step predictions,
- and a markedly higher R^2 score.

The GRU performs strongly on the raw bias, but the LSTM benefits noticeably from the additional structure introduced by the decomposition. Overall, the experiments show that

modelling the residual and drift separately is an effective strategy for high-precision GNSS clock prediction.

V. Future Scope

Although this study focused on clock bias of a single satellite (AS-C12) for forecasting, the general process can easily be extended to multiple satellites and/or other GNSS constellations. As a logical next step, it should not be limited to one-step ahead forecasts, but multi-step ahead forecasts, either via prediction through recursion, or by testing the models that were developed for longer sequences. Another useful extension would be to add some basic physics of GNSS, satellite temperature behavior, or basic space weather indicators in order to add physical realism to improve the stability of the forecasts. A hybrid use of traditional methods, such as Kalman filtering, and neural networks may further enhance the capabilities to handle drift. Finally, it would beneficial for the utility of a system for applications such as PPP, precise timing, and monitoring of clock behavior in practice or operational use, to make the current implementation into a light-weight, user-friendliness service (e.g., via FastAPI) and uncertainty estimates.

VI. Conclusion

This study explored two different neural network approaches for forecasting GNSS satellite clock bias using precise IGS clock data from the CDDIS archive. In the first approach GRU was implemented directly on raw clock bias. The second approach used an LSTM trained on a trend–residual decomposition of the bias.

The GRU model performed well and achieved picosecond-level accuracy, but its error varied throughout the day, especially when the bias exhibited stronger drift. This behaviour showed that learning directly from the raw bias can be challenging because the model must capture both slow long-term drift and small short-term fluctuations simultaneously.

The LSTM approach addressed this issue by removing the linear trend before training. After reconstructing the full bias, the model achieved an R^2 of about 0.99 on the unseen test day, outperforming the GRU model.

Overall, the results confirm that satellite clock bias can be forecasted reliably using compact neural architectures, and that even minimal preprocessing—such as separating trend from residual can improve performance.

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